

Diagnosis information PDK control unit

Distance sensor information - shift rod 4 outside validity range

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Distance sensor 4 faulty

Fault setting condition(s)

- Distance sensor 4 outside valid value range (10 - 90%)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Hold last gear
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

Fault finding notes

- ♦ Check vehicle wiring harness and internal wiring harness
- ♦ Check distance sensor by moving the shift rods

P1735

Diagnosis information PDK control unit

Distance sensor information for shift rod 1/2/3/4, short-term fault

Diagnostic conditions

- Driving in all gears
- Sensor supply voltage OK = 5 V

Fault setting condition(s)

- Distance sensor 1/2/3/4 outside of valid value range (10 - 90%)

Fault effects

- ♦ Hold last gear
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No Start/Stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod

Fault finding notes

- ♦ Fault codes **P1735**, **P1736**, **P1737**, **P1738** are provided for information purposes only if they do not occur often

INFORMATION

- ♦ If fault codes **P1735**, **P1736**, **P1737**, **P1738** occur frequently and can be reproduced, the following applies:

- ♦ Check vehicle wiring harness and internal wiring harness
- ♦ Check distance sensor by moving the shift rods

Possible fault causes

If they occur often or together with one of the following fault codes: P1731, P1732, P1733, P1734

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground

- ◆ Loose contact or corrosion on connectors or lines
- ◆ Distance sensor 1/2/3/4 faulty

P1736

Diagnosis information PDK control unit

Distance sensor information for shift rod 1/2/3/4, short-term fault

Diagnostic conditions

- Driving in all gears
- Sensor supply voltage OK = 5 V

Fault setting condition(s)

- Distance sensor 1/2/3/4 outside of valid value range (10 - 90%)

Fault effects

- ♦ Hold last gear
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No Start/Stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod

Fault finding notes

- ♦ Fault codes **P1735, P1736, P1737, P1738** are provided for information purposes only if they do not occur often

INFORMATION

- ♦ If fault codes **P1735, P1736, P1737, P1738** occur frequently and can be reproduced, the following applies:

- ♦ Check vehicle wiring harness and internal wiring harness
- ♦ Check distance sensor by moving the shift rods

Possible fault causes

If they occur often or together with one of the following fault codes: P1731, P1732, P1733, P1734

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground

- ◆ Loose contact or corrosion on connectors or lines
- ◆ Distance sensor 1/2/3/4 faulty

P1737

Diagnosis information PDK control unit

Distance sensor information for shift rod 1/2/3/4, short-term fault

Diagnostic conditions

- Driving in all gears
- Sensor supply voltage OK = 5 V

Fault setting condition(s)

- Distance sensor 1/2/3/4 outside of valid value range (10 - 90%)

Fault effects

- ♦ Hold last gear
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No Start/Stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod

Fault finding notes

- ♦ Fault codes **P1735, P1736, P1737, P1738** are provided for information purposes only if they do not occur often

INFORMATION

- ♦ If fault codes **P1735, P1736, P1737, P1738** occur frequently and can be reproduced, the following applies:

- ♦ Check vehicle wiring harness and internal wiring harness
- ♦ Check distance sensor by moving the shift rods

Possible fault causes

If they occur often or together with one of the following fault codes: P1731, P1732, P1733, P1734

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground

- ◆ Loose contact or corrosion on connectors or lines
- ◆ Distance sensor 1/2/3/4 faulty

P1738

Diagnosis information PDK control unit

Distance sensor information for shift rod 1/2/3/4, short-term fault

Diagnostic conditions

- Driving in all gears
- Sensor supply voltage OK = 5 V

Fault setting condition(s)

- Distance sensor 1/2/3/4 outside of valid value range (10 - 90%)

Fault effects

- ♦ Hold last gear
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No Start/Stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod

Fault finding notes

- ♦ Fault codes **P1735, P1736, P1737, P1738** are provided for information purposes only if they do not occur often

INFORMATION

- ♦ If fault codes **P1735, P1736, P1737, P1738** occur frequently and can be reproduced, the following applies:

- ♦ Check vehicle wiring harness and internal wiring harness
- ♦ Check distance sensor by moving the shift rods

Possible fault causes

If they occur often or together with one of the following fault codes: P1731, P1732, P1733, P1734

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground

- ◆ Loose contact or corrosion on connectors or lines
- ◆ Distance sensor 1/2/3/4 faulty

P173B

Diagnosis information PDK control unit

Transmission input shaft speed sensor, clutch 1 - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Transmission input shaft speed sensor of clutch 1 faulty

Fault setting condition(s)

- Sensor voltage < threshold value for short circuit to ground (0.15 V)
- Sensor voltage > threshold value for short circuit to B+ (4.5 V)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Electrical substitute value is assumed
- ♦ No start/stop

P173C

Diagnosis information PDK control unit

Transmission input shaft speed sensor, clutch 2 - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Transmission input shaft speed sensor of clutch 2 faulty

Fault setting condition(s)

- Sensor voltage < threshold value for short circuit to ground (0.15 V)
- Sensor voltage > threshold value for short circuit to B+ (4.5 V)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Electrical substitute value is assumed
- ♦ No start/stop

P173D

Diagnosis information PDK control unit

Transmission input shaft speed sensor, clutch 1 - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Transmission input shaft speed sensor of clutch 1 faulty

Fault setting condition(s)

- Sensor voltage < threshold value for short circuit to ground (0.15 V)
- Sensor voltage > threshold value for short circuit to B+ (4.5 V)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Electrical substitute value is assumed
- ♦ No start/stop

P173E

Diagnosis information PDK control unit

Transmission input shaft speed sensor, clutch 2 - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Transmission input shaft speed sensor of clutch 2 faulty

Fault setting condition(s)

- Sensor voltage < threshold value for short circuit to ground (0.15 V)
- Sensor voltage > threshold value for short circuit to B+ (4.5 V)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Electrical substitute value is assumed
- ♦ No start/stop

P1743

Diagnosis information PDK control unit

Transmission input shaft speed sensor plausibility check, clutch 1

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Transmission input shaft speed sensor faulty
- ♦ Slipping clutch

Fault setting condition(s)

- Comparison with the engine speed

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Electrical substitute value is assumed
- ♦ No start/stop

P1744

Diagnosis information PDK control unit

Transmission input shaft speed sensor plausibility check, clutch 2

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Transmission input shaft speed sensor faulty
- ♦ Slipping clutch

Fault setting condition(s)

- Comparison with the engine speed

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Electrical substitute value is assumed
- ♦ No start/stop

P1745

Diagnosis information PDK control unit

Input shaft overspeed threshold B

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Speed sensors faulty

Fault setting condition(s)

- The fault is detected when the speed of speed sensor 1 or 2 exceeds the upper threshold.
- The fault is healed when the speed of speed sensor 1 or 2 falls below the lower threshold or when a corresponding gear is disengaged or a new, higher gear is engaged in the gear train half.

Fault effects

- ♦ Clutches 1 and 2 are opened
- ♦ All gears in the transmission to neutral
- ♦ Clutch is open when the speed is < 4,000 rpm or when a corresponding gear is disengaged or a new, higher gear is engaged in the gear train half.

Fault finding notes

- ♦ Check the wiring harness and the corresponding component
- ♦ Erase fault memory and carry out a test drive

P1749

Diagnosis information PDK control unit

Variant coding not valid

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Incorrect program and data versions programmed
- ♦ Wrong coding in the vehicle data

Fault setting condition(s)

- Variant ID = 0

Fault effects

- ♦ Engine malfunction message is displayed

Fault finding notes

- ♦ Update of the correct variant code according to country identifier
- ♦ Variant must be written with the Tester

P174B

Diagnosis information PDK control unit

Invalid allocation of control unit and transmission hardware (interchanged installation)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ The control unit is installed in the wrong vehicle

Fault setting condition(s)

- The vehicle identification in the control unit does not match the vehicle identification number on the CAN

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Clutches are open
- ♦ Adaptation values of the transmission are set to invalid

Fault finding notes

- ♦ Perform adaptation of the shift rods and clutches

P174C

Diagnostic information - PDK control unit

Calibration data do not match the SW version

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Incorrect or missing adaptation values
- ♦ Wrong or faulty data record

Fault setting condition(s)

- Adaptation values do not match the software version

Fault effects

- ♦ Transmission fault display in instrument cluster control unit
- ♦ Emergency stop function
- ♦ All hydraulic control elements are switched off
- ♦ All low-side electric pressure adjusters are switched off
- ♦ All low-side solenoid valves are switched off
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ Ban one-touch operation and Sport program
- ♦ Start enable only in position P
- ♦ Electric pressure adjusters 5 and 6 are depressurized and multiplexer is changed over to locked shift rod
- ♦ Pressure control is switched off

Fault finding notes

- ♦ Reprogram PDK control unit and perform **calibration after part replacement**

P174E

Diagnosis information PDK control unit

Data record invalid

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Incorrect or faulty data version
- ♦ PDK control unit faulty

Fault setting condition(s)

- Internal diagnostic routine is faulty

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Emergency stop function

Diagnosis information PDK control unit

Supply voltage

Diagnostic conditions

- Ignition on
- with overvoltage Engine running

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Generator faulty
- ♦ Battery faulty

Fault setting condition(s)

- Supply voltage < 7 V
- Supply voltage > 16.5 V
- Supply voltage > 7... 9 V

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ All pressure regulators and solenoid valves are switched off
- ♦ Hydraulic control is switched off
- ♦ Gear held
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ No emergency program diagnostics
- ♦ One-touch operation and Sport program are deactivated
- ♦ Start enable possible only in position P
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ Pressure-sensor plausibility check is switched off
- ♦ Pressure control is switched off
- ♦ Stop message from CAN is contradicted

- ◆ Substitute wheel speed

Control unit setpoints

- ◆ Supply voltage at terminal 30 $9.1\text{ V} \Rightarrow 16.4\text{ V}$
- ◆ Voltage at terminal 15 $9.5\text{ V} \Rightarrow 16.0\text{ V}$

P1757

Diagnosis information PDK control unit

Input shaft 1 overspeed; 2nd step

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Clutch 1 faulty

Fault setting condition(s)

- The fault is detected when the speed of input shaft 1 exceeds the upper threshold (2nd step).

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Driving with gear train half 2
- ♦ Clutches 1 and 2 are opened
- ♦ Low-side electrical pressure adjuster 1 is switched off
- ♦ Only 2nd, 4th and 6th gears are possible
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

Diagnosis information PDK control unit

Input shaft 2 overspeed; 2nd step

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Clutch 2 faulty

Fault setting condition(s)

- The fault is detected when the speed of input shaft 2 exceeds the upper threshold (2nd step).

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Driving with gear train half 1
- ♦ Clutches 1 and 2 are opened
- ♦ Low-side electrical pressure adjuster 2 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P1760

Diagnosis information PDK control unit

Double fault: two faults demand alternative measures with equal priority rating

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Two additional faults of the same priority are entered

Fault setting condition(s)

- Inadmissible combination of alternative measures

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Emergency stop function
- ♦ All hydraulic control elements are switched off
- ♦ All low-side electrical pressure adjusters are switched off
- ♦ Low-side solenoid valves are switched off
- ♦ Gear held
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ One-touch operation and Sport program are deactivated
- ♦ Start enable only in position P
- ♦ Electrical pressure actuators 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ Pressure-sensor plausibility check is switched off
- ♦ Pressure control is switched off

P1764

Diagnosis information PDK control unit

Gear engagement lock

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Gear engagement lock

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit

P1765

Diagnosis information PDK control unit

Gear disengagement lock

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Gear disengagement lock

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Torque limitation
- ♦ Moving off in all gears
- ♦ Top speed limitation via engine intervention (lowest engaged gear)

P1766

Diagnosis information PDK control unit

Gear skip

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Gear skip

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit

P1767

Diagnosis information PDK control unit

Multiplexer error

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Software faulty

Fault setting condition(s)

- Multiplexer error

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Electrical pressure actuators 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod

Fault finding notes

- ♦ Install new software

P1771

Diagnosis information PDK control unit

Gear valve 1 stuck on pressure, minus or multiplexer fault

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Gear valve 1 stuck on pressure
- Gear valve 1 stuck on minus or multiplexer error

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Torque limitation
- ♦ Moving off in all gears
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Only 1st, 2nd, 4th and 7th gears are possible

P1772

Diagnosis information PDK control unit

Gear valve 2 stuck on pressure, minus or multiplexer fault

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Gear valve 2 stuck on pressure
- Gear valve 2 stuck on minus or multiplexer error

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Torque limitation
- ♦ Moving off in all gears
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Only 3rd, 5th, 6th and R gears are possible

P1773

Diagnosis information PDK control unit

Gear valve 1 stuck on pressure, minus or multiplexer fault

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Gear valve 1 stuck on pressure
- Gear valve 1 stuck on minus or multiplexer error

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Torque limitation
- ♦ Moving off in all gears
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Only 1st, 2nd, 4th and 7th gears are possible

P1774

Diagnosis information PDK control unit

Gear valve 2 stuck on pressure, minus or multiplexer fault

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Gear valve 2 stuck on pressure
- Gear valve 2 stuck on minus or multiplexer error

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Torque limitation
- ♦ Moving off in all gears
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Only 3rd, 5th, 6th and R gears are possible

P17A1

Diagnostic information - PDK control unit

Electric pressure adjuster 1 (clutch 1)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 1 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary current of low-side pressure adjuster 1 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 1 is switched off
- ♦ Low-side electric pressure adjuster 1 is switched off
- ♦ Only 2nd, 4th and 6th gears are possible
- ♦ No R gear
- ♦ Moving off in 2nd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ♦ If short circuit to B+, then all hydraulic control elements are switched off

► Breakdown

P17A2

Diagnostic information - PDK control unit

Electric pressure adjuster 2 (clutch 2)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 2 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 2 not energized
- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

P17A3

Diagnostic information - PDK control unit

Electric pressure adjuster 3 (system pressure)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 3 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 3 not energized
- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown